Module 2
Random Variables

Random Experiment → The experiment whose future outcomes cannot be predicted in advance is called random experiment.

Sample Space → A collection of all possible outcomes.

Event → Subset of sample space S is called event.

I Exhaustive Events → The events $A_1, A_2, \ldots A_n$ of S.S. 'S' are said to be collectively exhaustive events of S.S. 'S' if they include all possible events i.e. $A_1 \cup A_2 \cup \ldots \cup A_n = S$.

Ex → ① Consider the tossing of coin then
$S = \{H, T\}$
& events $E_1 = \{H\}$ & $E_2 = \{T\}$ are exhaustive events.

② Consider the rolling of dice
$S = \{1, 2, 3, 4, 5, 6\}$.
$E_1 = \{$ getting an even no. $\} = \{2, 4, 6\}$.
$E_2 = \{$ getting an odd no. $\} = \{1, 3, 5\}$
E_1 & E_2 are collectively exhaustive events.

II Mutually exclusive events → The events $A_1, A_2, \ldots A_n$ are said to be mutually exclusive events if no two or more of them occur simultaneously in same experiment.

$$A_i \cap A_j = \phi, \ \forall\, i \neq j, \ i, j = 1, 2, \ldots n.$$

Ex (1) If a coin tossed $S = \{H, T\}$ then

$E_1 = \{H\}$ & $E_2 = \{T\}$ are mutually exclusive events.

(2) If a dice is rolled then $S = \{1, 2, 3, 4, 5, 6\}$

then $E_1 = \{2, 4\}$, $E_2 = \{1\}$, $E_3 = \{5, 6\}$.

are mutually exclusive events because

$E_1 \cap E_2 = E_2 \cap E_3 = E_1 \cap E_3 = \phi$.

(3) <u>Equally Likely Events</u>:- If one of the event can not be expected to happen in preference to another then such events are said to be equally likely.

(1) In tossing an unbiased coin head or tail are equally likely events.

(2) In rolling an unbiased dice, all six faces are equally likely to turn up.

<u>Classical Approach to probability</u>:- If an experiment consists of n mutually exclusive, collectively exhaustive & equally likely outcomes & an event A can occur in 'm' of these 'n' ways then

$$P(A) = \frac{m}{n} \quad \text{or} \quad P(A) = \frac{n(A)}{n(S)} = \frac{|A|}{|S|}$$

$$\text{or} \quad P(A) = \frac{\text{No. of cases favourable to A}}{\text{Exhaustive no. of cases in S}}$$

Ex $S = \{H, T\} \Rightarrow |S| = 2$

$A = \{H\} \Rightarrow |A| = 1$ then $P(A) = \frac{1}{2}$

Statistical Def^n of probability.

$n \rightarrow$ trails.

$A \rightarrow$ Expected event

n_A = No. of occurance of A

$$\boxed{P(A) = \lim_{n \to \infty} \frac{n_A}{n}}$$

$\Rightarrow$ **Axioms of probability** $\rightarrow$ Let S be S.S. & A be an event associated with random experiment Then P(A) satisfies the following axioms:

(i) $0 \le P(A) \le 1$.

(ii) $P(S) = 1$

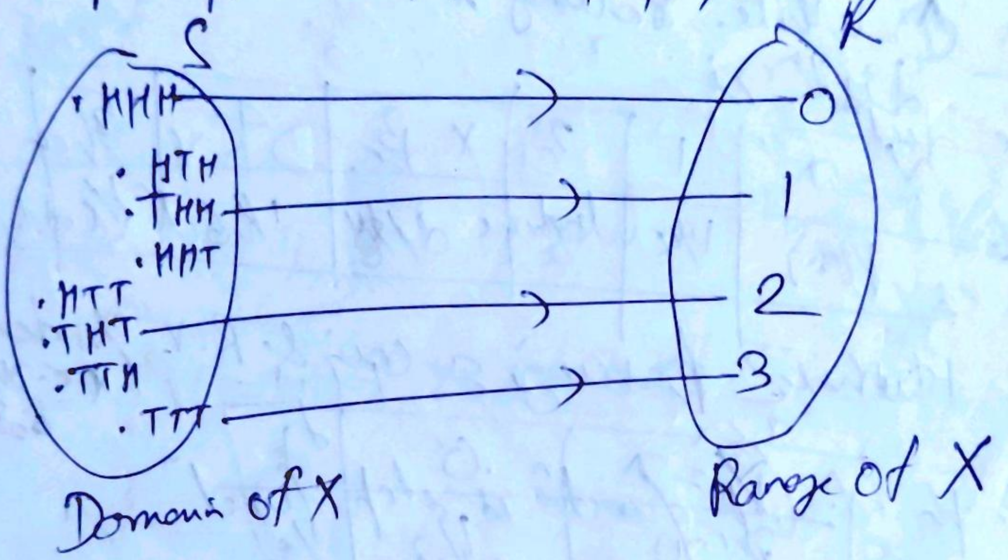
(iii) If A & B are mutually exclusive ($\rightarrow A \cap B = \phi$. & $P(A \cap B) = 0$) events in S then $P(A \cup B) = P(A) + P(B)$

Addition Th^m of Prob. $\rightarrow$ If A & B be events

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

⇒ **Random Variables** ⇒ A R.V. X is a function $X : S \to \mathbb{R}$ that assign a real no. $X(s)$ to each $s \in S$ (Sample space) corresponding to a random experiment E. Hence domain of R.V. is S, range is $(-\infty, \infty)$ & image may be subset of real number ($\mathbb{R}$).

Ex let a coin is tossed thrice & the R.V. X denotes the no. of tails that turn up. Hence $X = 0, 1, 2, 3$.



⇒ **Discrete R.V.** ⇒ A discrete R.V. has either finite or countably infinite no. of values.

Ex ① No. ~~of~~ shown when die is thrown.
② No. of white balls drawn from a bag containing 6 white & 2 black balls.

⇒ **Continuous R.V.** ⇒ A R.V. which can take infinite no. of values in an interval is known as continuous R.V.

Ex ① The height of person ② Weight of fish
③ $X = \{x \in \mathbb{R} : 0 < x < 1\}$.

→ probability distribution of D.R.V ⇒ The probability distribution of DRV lists all the possible values that the R.V. can assume & their corresponding probabilities. If a R.V. assumes the values $x_0, x_1, x_2, \ldots$ with probabilities $p_0, p_1, p_2, \ldots$ then prob. distⁿ is

$X = x$	x_0	x_1	x_2
$P(x)$	p_0	p_1	p_2

Ex ⇒ ① While rolling the fair die, the prob. distⁿ is

$X = x$	1	2	3	4	5	6
$P(x)$	1/6	1/6	1/6	1/6	1/6	1/6

② While tossing a coin & R.V. X denotes the no. of heads obtained.

$X = x$	0	1
$P(x)$	1/2	1/2

→ probability mass function ⇒ If X is DRV with distinct values $x_1, x_2, \ldots x_n, \ldots$ then the function $p(x)$ defined as

$$p_X(x) = \begin{cases} P(X = x_i) = p_i, & \text{if } x = x_i \\ 0, & \text{if } x \neq x_i, \ i = 1, 2, \ldots \end{cases}$$

is pmf of R.V. X must satisfy the following conditions:

(i) $p(x_i) \geq 0, \ \forall i$ (ii) $\sum_{i=1}^{\infty} p(x_i) = 1$

Q. Check whether the following functions serve as pmf

(1) $P(X=x) = \frac{x-2}{2}$, $\forall x = 1, 2, 3, 4$

(2) $P(X=x) = \frac{x^2}{25}$, $\forall x = 1, 2, 3, 4$.

Sol^n,

x	:	1	2	3	4
$P(X=x)$	:	$-1/2$	0	$1/2$	1

As $P(X=1) = -\frac{1}{2} < 0$, hence $P(X)$ is not a pmf.

(2)

x	:	1	2	3	4
$P(X=x)$	:	$1/25$	$4/25$	$9/25$	$16/25$

$P(X=x) > 0$, $\forall x = 1, 2, 3, 4$

but $\sum_{x=1}^{4} P(X=x) = \frac{30}{25} = \frac{6}{5} > 1$

Hence not pmf.

Q. Four bad oranges are mixed accidently with 16 good oranges. Find prob. dist^n of the no. of bad oranges in draw of two oranges.

Sol^n, Let the R.V. X denote the no. of bad oranges in a draw of two oranges.

Hence $X = 0, 1, 2$.

$P(X=0) =$ Prob. of getting 2 good oranges $= \frac{^{16}C_2}{^{20}C_2}$

$= \frac{12}{19}$

$P(X=1)$ = prob of getting 1 good orange & 1 bad orange $= \dfrac{{}^4C_1 \; {}^{16}C_1}{{}^{20}C_2}$

$= \dfrac{32}{95}$.

$P(X=2)$ = prob. of getting 2 bad oranges

$= \dfrac{{}^4C_2}{{}^{20}C_2} = \dfrac{3}{95}$

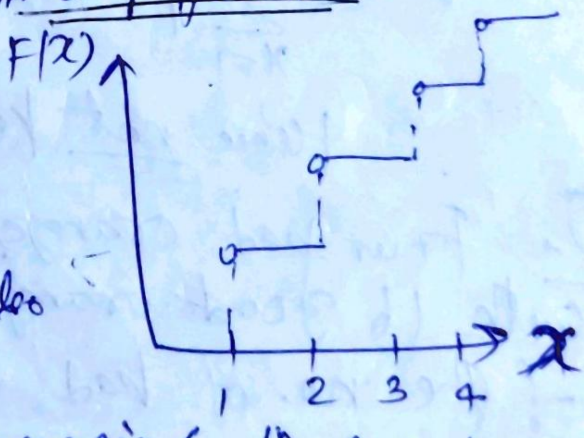
Hence prob. distn is

X :	0	1	2
$P(X)$:	$12/19$	$32/95$	$3/95$

Discrete distn functn or cdf → Let X be D.R.V. then its discrete distn functn or cumulative distn functn (cdf) is defined as

$$F(x) = \sum_{x_i \le x} p_i \qquad \text{i.e. } F(x_j) = P(X \le x_j) = \sum_{i \le j} p_i$$

The graph of distn functn is step functn



The distn functn $F(x)$ satisfies the following prop.

(i) $0 \le F(x) \le 1$, as $F(x)$ is also a probability

(ii) $F(x)$ is a monotonic non decreasing functn of x i.e. $x_1 \le x_2$ then $F(x_1) \le F(x_2)$.

Solve Ex 5.1 by S.C. Gupta P-5.6

On page 5.10; Ex 5.2.

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Q. A random variable X has the following probability distribution:

x	0	1	2	3	4	5	6	7
$p(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$

(i) Find k (ii) Evaluate $P(X<6)$, $P(X\geq 6)$ & $P(0<X<5)$.

(iii) If $P(X\leq c) > \frac{1}{2}$, find minimum value of c.

(iv) Determine the distribution function of X.

Sol: (i) $\because \sum_{x=0}^{7} p(x) = 1$

$$0+k+2k+2k+3k+k^2+2k^2+7k^2+k = 1$$

$$10k^2+9k-1=0$$

$$10k^2+10k-k-1=0$$

$$10k(k+1)-1(k+1)=0$$

$$k = \frac{1}{10}$$; & $k=-1$ is rejected $\because$ prob can't be -ve.

(ii) $P(X<6) = P(X=0)+P(X=1)+\dots+P(X=5)$

$$= \frac{1}{10}+\frac{2}{10}+\frac{2}{10}+\frac{3}{10}+\frac{1}{100} = \frac{81}{100}.$$

$$P(X\geq 6) = 1-P(X<6) = 1-\frac{81}{100} = \frac{19}{100}.$$

$$P(0<X<5) = P(X=1)+P(X=2)+P(X=3)+P(X=4)$$

$$= 8k = \frac{4}{5}.$$

(iii) $P(X\leq c) > \frac{1}{2}$, By trial, we get $c=4$.

(iv)

X	$F_X(x) = P(X\leq x)$
0	0
1	$k = 1/10$
2	$3k = 3/10$
3	$5k = 5/10$
4	$8k = 4/5$
5	$8k+k^2 = 81/100$
6	$8k+3k^2 = 83/100$
7	$9k+10k^2 = 1$.

→ Prob. distⁿ of continuous R.V. → Continuous (36).

Prob. distⁿ of x by the funct $f(x)$ s.t. prob of x lying in small interval $\left(x-\frac{dx}{2}, x+\frac{dx}{2}\right)$ is $f(x)\,dx$ i.e.

$$P\left[x-\frac{dx}{2} < X < x+\frac{dx}{2}\right) = f(x)\,dx.$$

The continuous curve $y=f(x)$ is known as prob. curve.

Prob. density function → The function $f(x)$ for a CRV X is said to be pdf if

(i) $f(x) \geq 0$; $-\infty < x < \infty$

(ii) $\int_{-\infty}^{\infty} f(x)\,dx = 1.$

Moreover $P(a \leq X \leq b) = \int_a^b f(x)\,dx.$

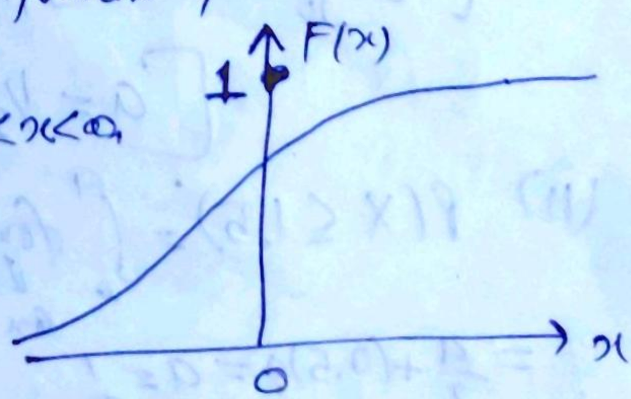
Remark → When X is a CRV then $P(a \leq X \leq b)$

$= P(a \leq X < b) = P(a < X \leq b) = P(a < X < b)$

because $P(X=a) = P(a \leq X \leq a) = \int_a^a f(x)\,dx = 0.$

The Distribution Function or cdf → Let X be a CRV with pdf $f(x)$ then the function $F_X(x)$ is called continuous distⁿ function or cumulative distⁿ function (cdf) of RV, X where

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f(x)\,dx, \quad -\infty < x < \infty.$$



Properties → (i) $0 \leq F(x) \leq 1$, $-\infty < x < \infty$

(ii) $F(x) = \int_{-\infty}^{x} f(x)\,dx \Rightarrow F'(x) = f(x)$

i.e. $\frac{d}{dx}F(x) = f(x) \geq 0.$

Hence $F(x)$ is non-decreasing function of x.

(iii) $F(-\infty) = \lim_{x \to -\infty} F(x) = \int_{-\infty}^{-\infty} f(x)\,dx = 0$

$F(+\infty) = \lim_{x \to \infty} F(x) = \int_{-\infty}^{\infty} f(x)\,dx = 1.$

(iv) $P(a \le X \le b) = \int_a^b f(x)\,dx = \int_{-\infty}^{b} f(x)\,dx - \int_{-\infty}^{a} f(x)\,dx$

$= P(X \le b) - P(X \le a)$

$P(a \le X \le b) = F(b) - F(a)$

Y.McGraw P-2.14

Q) Let X be a C.R.V. with p.d.f.

$$f(x) = \begin{cases} ax & , 0 \le x \le 1 \\ a & , 1 \le x \le 2 \\ -ax+3a & , 2 \le x \le 3. \\ 0 & , \text{elsewhere} \end{cases}$$

(i) Determine the constant a.

(ii) Find $P(X \le 1.5)$

(iii) Determine the cdf & hence find $P(X \le 2.5)$

Sol^n (i) As $f(x)$ is given to be pdf, hence

$$\int_0^{\infty} f(x)\,dx = 1 \Rightarrow \int_0^1 \underset{ax}{\underbrace{f(x)}}\,dx + \int_1^2 \underset{a}{\underbrace{f(x)}}\,dx + \int_2^3 \underset{(-ax+3a)}{\underbrace{f(x)}}\,dx = 1$$

$$a\left[\frac{x^2}{2}\right]_0^1 + a[x]_1^2 + \left[-a\left\{\frac{x^2}{2}\right\} + 3ax\right]_2^3 = 1.$$

$$\boxed{a = 1/2}$$

(ii) $P(X \le 1.5) = \int_0^1 \underset{ax}{\underbrace{f(x)}}\,dx + \int_1^{1.5} \underset{a}{\underbrace{f(x)}}\,dx = a\left[\frac{x^2}{2}\right]_0^1 + [ax]_1^{1.5}$

$= \frac{a}{2} + (0.5)a = a = \frac{1}{2}.$

(iii) For $x<0$ $\quad F(x)=0$.

For $0 \le x \le 1$, $F(x) = \int_0^x f(x)dx = \int_0^x ax\,dx = a\left[\frac{x^2}{2}\right]_0^x$

$$= \frac{ax^2}{2} = \frac{x^2}{4} \quad \left\{\because a = \frac{1}{2}\right\}$$

For $1 \le x \le 2$, $F(x) = \int_0^1 f(x)dx + \int_1^x f(x)dx$.

$$= \int_0^1 ax\,dx + \int_1^x a\,dx = \frac{x}{2} - \frac{1}{4}.$$

For $2 \le x \le 3$, $F(x) = \int_0^1 f(x)dx + \int_1^2 f(x)dx + \int_2^x f(x)dx$

$$= \int_0^1 ax\,dx + \int_1^2 a\,dx + \int_2^x (-ax+3a)dx$$

$$= \frac{-5}{4} + \frac{3x}{2} - \frac{x^2}{4}.$$

For $x \ge 3$, $F(x) = \int_0^\infty f(x)dx = 1$.

Hence required cdf is

$$F(x) = \begin{cases} 0, & x<0 \\ x^2/4, & 0 \le x \le 1 \\ \frac{x}{2} - \frac{1}{4}, & 1 \le x \le 2 \\ \frac{-x^2}{2} + \frac{3x}{2} - \frac{5}{4}, & 2 \le x \le 3 \\ 1, & x \ge 3. \end{cases}$$

Now $P(X \le 2.5) = -\frac{(2.5)^2}{2} + \frac{3}{2}(2.5) - \frac{5}{4}$

$$= 0.9375.$$

p 36.2 S C Gupta (39)

⇒ Mathematical Expectation of a R.V. → or expected value of R.V. →

$$\boxed{E(X) = \sum_x x f(x)}$$ For D.R.V.

where $f(x)$ is pmf.

$$\boxed{E(X) = \int_{-\infty}^{\infty} x f(x)\,dx}$$ for C.R.V.

where $f(x)$ is pdf.

provided the RHS of above series or integral is absolutely convergent.

⇒ Expected value of function of a R.V. →

Let $g(X)$ be a function on a R.V. X then

$$E[g(X)] = \sum_x g(x) f(x) \quad \text{For D.R.V.}$$

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x)\,dx \quad \text{For C.R.V.}$$

Choose $g(X) = X$

$$\boxed{\left.\begin{aligned} E(X) &= \sum_{i=1}^{n} x_i f(x_i) \quad \text{DRV} \\ E(X) &= \int_{-\infty}^{\infty} x f(x)\,dx \quad \text{CRV} \end{aligned}\right\}}$$ → Mean of X → $\mu_X = E(X)$

Choose $g(X) = X.X = X^2$

$$E(X^2) = \begin{cases} \sum_{i=1}^{n} x_i^2 f(x_i), & \text{DRV} \\ \int_{-\infty}^{\infty} x^2 f(x)\,dx, & \text{CRV} \end{cases}$$

$$Var(X) = E(X^2) - [E(X)]^2 = E[X - E(X)]^2, \quad SD = +\sqrt{Var(X)}$$ ↑ + sign

$\sigma_X^2 = Var(X)$

$Var(X) = \sigma_x^2 = E[X - E(X)]^2 = E[X - \mu_X]^2$

$= \sum_i (x_i - \mu_X)^2 f(x_i)$, if X is DRV

$= \int_{-\infty}^{\infty} (x - \mu_X)^2 f(x)\,dx$ if X is CR.V.

$$\boxed{Var(X) = E(X^2) - [E(X)]^2}$$

$S.D = +\sqrt{Var(X)} = \sigma_x$

Properties ① $E(aX) = aE(X)$; a is any constant.

② $Var(aX) = a^2 Var(X)$ or $Var(aX+b) = a^2 Var(X)$

Proof: Let $Y = aX + b$ then $E(Y) = aE(X) + b$.

Subtracting, $Y - E(Y) = a[X - E(X)]$

squaring & taking expectation of both sides, we get

$E[Y - E(Y)]^2 = a^2 E[X - E(X)]^2$

$Var(Y) = a^2 Var(X)$

i.e. $\boxed{Var(aX+b) = a^2 Var(X)}$

③ $E(X+Y) = E(X) + E(Y)$

④ If X & Y are independent RVs then

$E(XY) = E(X)E(Y)$

1.6.11 SCG

Q. Let X be a R.V. with following prob dist?

x:	-3	6	9
P(X=x):	1/6	1/2	1/3

Find $E(X)$, $E(X^2)$ & using the law of expectation, evaluate $E(2X+1)^2$.

Sol: $E(X) = \sum x\,p(x) = (-3)\cdot\frac{1}{6} + 6\cdot\frac{1}{2} + 9\cdot\frac{1}{3} = \frac{11}{2}$

$E(X^2) = \sum x^2 p(x) = 9 \times \frac{1}{6} + 36 \times \frac{1}{2} + 81 \times \frac{1}{3} = \frac{93}{2}$.

$\therefore E(2X+1)^2 = E(4X^2 + 4X + 1) = 4E(X^2) + 4E(X) + 1$

$= 4 \times \frac{93}{2} + 4 \times \frac{11}{2} + 1 = 209$

Two dimensional R.V.s. Let X & Y be two R.V. defined on sample space S, then the function (X, Y) that assigns a point in $\mathbb{R}^2 (= \mathbb{R} \times \mathbb{R})$ is called a two dimensional R.V.

If the possible values of (X, Y) are finite or countably infinite, (X, Y) is called a two-dim Discrete RV. When (X, Y) is a two dim DRV the possible values of RV. may be represented as (x_i, y_j); $i = 1, 2, \ldots m; \ldots$ $j = 1, 2, \ldots n, \ldots$

If (X, Y) can assume all values in a specified region R in the xy-plane, is called a two dim continuous RV.

Probability function of (X, Y) ⟹ If (X, Y) be a two-dim DRV s.t. $P(X = x_i, Y = y_j) = p_{ij}$ then p_{ij} is called probability mass function (pmf) or simply probability function of (X, Y) provided the following conditions are satisfied:

(i) $p_{ij} \geq 0, \forall i \,\&\, j$

(ii) $\sum_j \sum_i p_{ij} = 1$.

The set of ~~triples~~ triples $\{x_i, y_j; p_{ij}\}$, $i = 1, 2, \ldots m, \ldots$ $j = 1, 2, \ldots n, \ldots$ is called the joint prob. dist of (X, Y).

The prob that point (x,y) will lie in infinitesimal rectangular region of area $dx\,dy$ is given by

⇒ **Joint prob. density function** → If (X,Y) is a two dim CRV s.t.

$$P\left\{x-\frac{dx}{2} \le X \le x+\frac{dx}{2} \ \&\ y-\frac{dy}{2} \le Y \le y+\frac{dy}{2}\right\}$$
$$= f(x,y)\,dx\,dy$$

then $f(x,y)$ is called the joint pdf of (X,Y), provided $f(x,y)$ satisfies the following conditions:

(i) $f(x,y) \geq 0,\ \forall (x,y) \in R$, where R is range space

(ii) $\displaystyle\iint_R f(x,y)\,dx\,dy = 1.$

Moreover if D is a subspace of range space R, $P\{(X,Y) \in D\}$ is defined as:

$$P\{(X,Y) \in D\} = \iint_D f(x,y)\,dx\,dy.$$

In particular

$$P\{a \le X \le b,\ c \le Y \le d\} = \int_c^d \int_a^b f(x,y)\,dx\,dy$$

Cumulative dist^n function → If (X,Y) be two-dim RV (discrete or continuous) then $F(x,y) = P\{X \le x \ \&\ Y \le y\}$ is called the cdf of (X,Y).

$$F(x,y) = \sum_{\substack{j \\ y_j \le y}} \sum_{\substack{i \\ x_i \le x}} p_{ij} \qquad \text{Discrete RV}$$

$$= \int_{-\infty}^{y} \int_{-\infty}^{x} f(x,y)\,dx\,dy, \qquad \text{CRV.}$$

Properties → (i) $F(-\infty, y) = 0 = F(x,-\infty)$ & $F(\infty,\infty) = 1$.

(ii) $P\{a < X < b,\ Y \le y\} = F(b,y) - F(a,y)$

(iii) $P\{X \le x,\ c < Y < d\} = F(x,d) - F(x,c)$

(iv) $P\{a < X < b, c < Y < d\} = F(b,d) - F(a,d) - F(b,c) + F(a,c)$.

(v) At points of continuity of $f(x,y)$

$$\frac{\partial^2 F}{\partial x \partial y} = f(x,y).$$

⇒ Marginal prob. distⁿ:

$P(X=x_i) = P\{(X=x_i \ \& \ Y=y_1)$ or $(X=x_i \ \& \ Y=y_2)$ or etc$\}$.

$$= p_{i1} + p_{i2} + \dots = \sum_j p_{ij} = p_{i*}$$

$P(X=x_i) = \sum_j p_{ij}$ is called the marginal prob. function of X. It is defined for $X = x_1, x_2, \dots$ & denoted as p_{i*}. The collection of pairs $\{x_i, p_{i*}\}$, $i = 1,2,3,\dots$ is called the marginal probability distribution of X.

Similarly the collection of pairs $\{y_j, p_{*j}\}$, $j=1,2,3\dots$ is called marginal prob. distⁿ of X where

$$p_{*j} = \sum_i p_{ij} = P(Y=y_j)$$

In continuous case,

$$P\{x - \tfrac{dx}{2} \le X \le x + \tfrac{dx}{2} ; -\infty < Y < \infty\}.$$

$$= \int_{-\infty}^{\infty} \int_{x-\frac{dx}{2}}^{x+\frac{dx}{2}} f(x,y)\,dx\,dy.$$

$$= \left[\int_{-\infty}^{\infty} f(x,y)\,dy\right] dx \quad \{\text{since } f(x,y) \text{ may be treated a constant in } (x-\tfrac{dx}{2}, x+\tfrac{dx}{2})\}$$

$= f_X(x)\,dx$. (say).

~~Similarly~~ $f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$ is called the marginal density of X.

Similarly, $f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$ is called the marginal density of Y.

Note: $P(a \le X \le b) = P(a \le X \le b, -\infty < Y < \infty)$

$$= \int_{-\infty}^{\infty}\int_{a}^{b} f(x,y)\,dx\,dy$$

$$= \int_{a}^{b}\left[\int_{-\infty}^{\infty} f(x,y)\,dy\right]dx$$

$$= \int_{a}^{b} f_X(x)\,dx.$$

Similarly, $P(c \le Y \le d) = \int_{c}^{d} f_Y(y)\,dy$.

⇒ Conditional prob. distⁿ: Let (X, Y) be DRV.

$$P\{X = x_i \mid Y = y_j\} = \frac{P\{X = x_i, Y = y_j\}}{P\{Y = y_j\}} = \frac{p_{ij}}{p_{*j}}$$

is called the conditional prob. functⁿ of X, given $Y = y_j$.

The collection of pairs $\left\{x_i, \frac{p_{ij}}{p_{*j}}\right\}$, $i = 1, 2, 3, \ldots$

is called the conditional prob distⁿ of X, given $Y = y_j$.

Similarly, the collection of pairs $\left\{y_j, \frac{p_{ij}}{p_{i*}}\right\}$, $j = 1, 2, 3, \ldots$ is called the conditional prob distⁿ of Y given $X = x_i$.

In continuous case,

$$P\left\{x-\frac{dx}{2} \le X \le x+\frac{dx}{2} \,\middle|\, Y=y\right\}$$

$$= P\left\{x-\frac{dx}{2} \le X \le x+\frac{dx}{2} \,\middle|\, y-\frac{dy}{2} \le Y \le y+\frac{dy}{2}\right\}$$

$$= \frac{f(x,y)\,dx\,dy}{f_Y(y)\,dy} = \left\{\frac{f(x,y)}{f_Y(y)}\right\} dx.$$

$\frac{f(x,y)}{f_Y(y)}$ is called the conditional density of X, given Y & is denoted by $f(x/y)$ or $f(x|y)$.

Similarly $\frac{f(x,y)}{f_X(x)}$ is called the conditional density of Y, given X. & is denoted by $f(y/x)$ or $f(y|x)$.

Independent RVs → If (X,Y) is two dim DRV s.t. $P\{X=x_i / Y=y_j\} = P(X=x_i)$ i.e. $\frac{p_{ij}}{p_{*j}} = p_{i*}$

i.e. $p_{ij} = p_{i*} \times p_{*j}$ $\forall\, i,j$ then X & Y are said to be independent RVs.

Q. The joint pmf of (X,Y) is given by $p(x,y) = k(2x+3y)$, $x=0,1,2$; $y=1,2,3$. Find all the marginal & conditional prob. distⁿ. Also find the prob. distⁿ of $(X+Y)$.

Sol^n: The joint prob. dist^n of (X, Y) is given below.

X \ Y	1	2	3
0	3k	6k	9k
1	5k	8k	11k
2	7k	10k	13k

$$\sum_{j=1}^{3}\sum_{i=0}^{2} p(x_i, y_j) = 1.$$

i.e. sum of all probabilities in above table is equal to 1.

i.e. $72k = 1$

$k = 1/72$.

Marginal Prob. dist^n of X: $\{i, p_{i*}\}$

$X = i$	$p_{i*} = \sum_{j=1}^{3} p_{ij}$
0	$p_{01} + p_{02} + p_{03} = 3k + 6k + 9k = \frac{18}{72}$
1	$p_{11} + p_{12} + p_{13} = 5k + 8k + 11k = 24k = \frac{24}{72}$
2	$p_{21} + p_{22} + p_{23} = 7k + 10k + 13k = 30k = \frac{30}{72}$.
	Total $= 1$.

Marginal prob. dist^n of X: $\{j, p_{*j}\}$

$Y = j$	$p_{*j} = \sum_{i=0}^{2} p_{ij}$
1	$p_{01} + p_{11} + p_{21} = 3k + 5k + 7k = 15k = \frac{15}{72}$
2	$p_{02} + p_{12} + p_{22} = 6k + 8k + 10k = \frac{24}{72}$
3	$p_{03} + p_{13} + p_{23} = 9k + 11k + 13k = \frac{33}{72}$
	Total $= 1$.

Conditional distribution of X, given Y = 1, is given by $\{i, P(X=i/Y=1)\} = \{i, P(X=i, Y=1)/P(Y=1)\}$

$$= \{i, p_{i1}/p_{*1}\}; i = 0,1,2.$$

$X=i$	p_{i1}/p_{*1}
0	$3k/15k = 1/5$
1	$5k/15k = 1/3$
2	$7k/15k = 7/15$
	Total = 1.

The other conditional dist^n are given below.

CPD of X given Y = 2

$X=i$	p_{i2}/p_{*2}
0	$6k/24k = 1/4$
1	$8k/24k = 1/3$
2	$10k/24k = 5/12$
	Total = 1.

CPD of X given Y = 3

$X=i$	p_{i3}/p_{*3}
0	$9k/33k = 3/11$
1	$11k/33k = 1/3$
2	$13k/33k = 13/33$
	Total = 1.

CPD of Y given X=0

$Y=j$	p_{0j}/p_{0*}
1	$3k/18k = 1/6$
2	$6k/18k = 1/3$
3	$9k/18k = 1/2$
	Total = 1.

CPD of Y given X=1

$Y=j$	p_{1j}/p_{1*}
1	$5k/24k = 5/24$
2	$8k/24k = 1/3$
3	$11k/24k = 11/24$
	Total = 1.

CPD of Y given X=2

$Y=j$	p_{2j}/p_{2*}
1	$7k/30k = 7/30$
2	$10k/30k = 1/3$
3	$13k/30k = 13/30$
	Total = 1.

Now Probability distribution of $(X+Y)$

$(X+Y)$	P
1	$p_{01} = 3/72$
2	$p_{02} + p_{11} = 11/72$
3	$p_{03} + p_{12} + p_{21} = 24/72$
4	$p_{13} + p_{22} = 21/72$
5	$p_{23} = 13/72$
	Total = 1

Q1 The joint pdf of a two-dim RV (X, Y) is given by $f(x,y) = xy^2 + \frac{x^2}{8}$, $0 \le x \le 2$, $0 \le y \le 1$. Compute $P(X>1)$, $P(Y<\frac{1}{2})$, $P(X>1 / Y<1/2)$, $P(Y<\frac{1}{2} / X>1)$, $P(X<Y)$ & $P(X+Y \le 1)$.

Proof: (i) $P(X>1) = \iint_{x>1} f(x,y)\, dx\, dy$

$$= \int_0^1 \int_1^2 \left(xy^2 + \frac{x^2}{8}\right) dx\, dy = \frac{19}{24}$$

(ii) $P(Y<1/2) = \iint_{y<\frac{1}{2}} \left(xy^2 + \frac{x^2}{8}\right) dx\, dy$

$$= \int_0^{1/2} \int_0^2 \left(xy^2 + \frac{x^2}{8}\right) dx\, dy = \frac{1}{4}$$

comma

(iii) $P(X>1, Y<1/2) = \int_{R_3} \left(xy^2 + \frac{x^2}{8}\right) dx\, dy$

$(x>1\ \&\ y<1/2)$

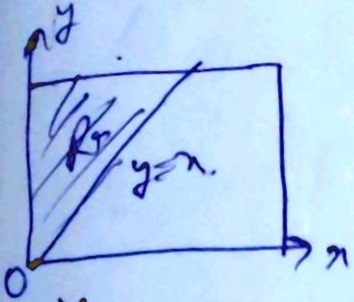
$$= \int_0^{1/2} \int_1^2 \left(xy^2 + \frac{x^2}{8}\right) dx\, dy.$$

$= 5/24.$

(iv) $P(X>1 / Y<1/2) = \dfrac{P(X>1, Y<1/2)}{P(Y<\frac{1}{2})} = \dfrac{5/24}{1/4} = 5/6.$

(v) $P(Y<\frac{1}{2} / X>1) = \dfrac{P(X>1, Y<1/2)}{P(X>1)} = \dfrac{5/24}{19/24} = \dfrac{5}{19}$

(vi) $P(X<Y) = \iint\limits_{\substack{R_4 \\ (x<y)}} \left(xy^2 + \frac{x^2}{8}\right) dx\,dy$



$$= \int_0^1 \int_0^y \left(xy^2 + \frac{x^2}{8}\right) dx\,dy = \frac{53}{480}.$$

(vii) $P(X+Y \leq 1) = \iint\limits_{\substack{R_5 \\ x+y\leq 1}} \left(xy^2 + \frac{x^2}{8}\right) dx\,dy$

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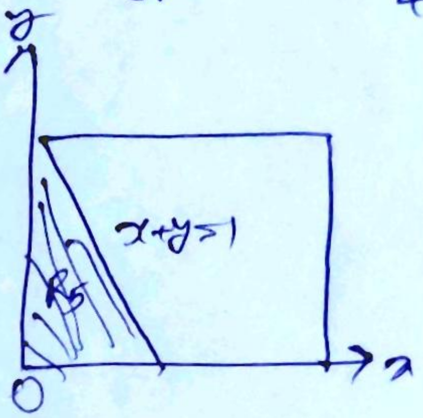
$$= \int_0^1 \int_0^{1-y} \left(xy^2 + \frac{x^2}{8}\right) dx\,dy = \frac{13}{480}.$$



$\Rightarrow$ Covariance $\rightarrow$ As the variance $E\{X-E(X)\}^2$ measures the variances of the R.V. X from its mean value $E(X)$, the quantity $E\{[X-E(X)][Y-E(Y)]\}$ measures the simultaneous variation of two R.V. X & Y from their respective means & hence it is called the covariance of X, Y & denoted as $Cov(X,Y)$.

$Cov(X,Y) = E\{[X-E(X)][Y-E(Y)]\}$ is also called the product moment of X & Y & is also denoted as $p(X,Y)$.

$$Cov(X,Y) = E(XY) - E(X)\cdot E(Y)$$

$$Cov(X,Y) = \frac{1}{n}\sum x_i y_i - \frac{1}{n}\sum x_i \cdot \frac{1}{n}\sum y_i$$

Q. Let X & Y be CRV with joint pdf

$$f_{X,Y}(x,y) = 3x \ , \quad 0 \le y \le x \le 1.$$
$$= 0 \ , \quad \text{otherwise}.$$

Find the co-variance between X & Y.

Sol^n The marginal pdfs, expectations & variances of X & Y are

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy = \int_0^x 3x\,dy = 3x^2 \ , \quad 0 \le x \le 1.$$

$$\Rightarrow E_{f_X}[X] = \int_{-\infty}^{\infty} x f_X(x)\,dx = \int_0^1 x \times 3x^2\,dx = \left[\frac{3}{4}x^4\right]_0^1 = \frac{3}{4}.$$

$$E_{f_X}[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x)\,dx = \int_0^1 x^2 \times 3x^2\,dx = \left[\frac{3}{5}x^5\right]_0^1 = \frac{3}{5},$$

$$\Rightarrow Var_{f_X}[X] = E_{f_X}[X^2] - \{E_{f_X}[X]\}^2 = \frac{3}{5} - \left(\frac{3}{4}\right)^2 = \frac{3}{80}.$$

Now $$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx = \int_y^1 3x\,dx = \left[\frac{3}{2}x^2\right]_y^1 = \frac{3}{2}(1-y^2),$$
$$0 \le y \le 1,$$

$$\Rightarrow E_{f_Y}[Y] = \int_{-\infty}^{\infty} y f_Y(y)\,dy = \int_0^1 y \times \frac{3}{2}(1-y^2)\,dy = \frac{3}{8}.$$

$$E_{f_X}[Y^2] = \int_{-\infty}^{\infty} y^2 f_Y(y)\,dy = \int_0^1 y^2 \times \frac{3}{2}(1-y^2)\,dy = \frac{1}{5}.$$

$\Rightarrow Var_{f_Y}[Y] = E_{f_Y}[Y^2] - \{E_{f_Y}[Y]\}^2 = E_{f_Y}[Y^2] - \{E_{f_Y}[Y]\}^2$ (53)

$$= \frac{1}{5} - \left(\frac{3}{8}\right)^2$$

$$= \frac{19}{320}.$$

& To compute the covariance we also need to compute $E_{f_{X,Y}}[XY]$.

$$E_{f_{X,Y}}[XY] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy\, f_{X,Y}(x,y)\, dy\, dx$$

$$= \int_0^1\int_0^x xy \times 3x\, dy . dx = \int_0^1 \left[\int_0^x y\,dy\right] 3x^2 dx$$

$$= \int_0^1 \left[\frac{y^2}{2}\right]_0^x 3x^2 dx = \int_0^1 \frac{x^2}{2} \times 3x^2 dx$$

$$= \frac{3}{2}\left[\frac{x^5}{5}\right]_0^1 = \frac{3}{10}.$$

$$\Rightarrow Cov_{f_{X,Y}}[X,Y] = E_{f_{X,Y}}[X,Y] - E_{f_X}[X]\, E_{f_Y}[Y]$$

$$= \frac{3}{10} - \frac{3}{4} \times \frac{3}{8} = \frac{3}{160}$$

$\Rightarrow$ <u>Moment Generating Function (MGF)</u> $\rightarrow$ MGF of RV X (discrete or continuous) is defined as $E(e^{tX})$, where t is a real variable & denoted as $M(t)$.

If X is discrete, then $M(t) = \sum_r e^{tx_r} p_r$.

where X takes the values $x_1, x_2, x_3 \dots$ with probabilities $p_1, p_2, p_3 \dots$

If X is a CRV with density function $f(x)$ then

$$M(t) = \int_{-\infty}^{\infty} e^{tx} f(x)\, dx.$$

<u>Properties of MGF</u> $\rightarrow$ ① $M(t) = \sum_{n=0}^{\infty} t^n E(X^n)/n!$

i.e. $E(X^n) = \mu_n'$ is the coeff. of $\frac{t^n}{n!}$ in the expansion of $M(t)$ in series of power of t.

② (part endg.) $\mu_n' = E(X^n) = \left[\frac{d^n}{dt^n} M(t)\right]_{t=0}$

③ If MGF of X is $M_X(t)$ & if $Y = aX+b$ then
$$M_Y(t) = e^{bt} M_X(at).$$

④ If X & Y are independent RV & $Z = X+Y$ then
$$M_Z(t) = M_X(t) M_Y(t).$$

Q→ If X represents the outcome, when a fair die is tossed, find the MGF of X & hence find E(X) & Var(X).

Sol^n The prob. dist^n of X is given by
$$p_i = p(X=i) = \frac{1}{6}, \quad i = 1, 2, \ldots 6.$$
$$M(t) = \sum_i e^{tx_i} p_i = \sum_{i=1}^{6} e^{ti} p_i$$
$$= \frac{1}{6}\left(e^t + e^{2t} + e^{3t} + e^{4t} + e^{5t} + e^{6t}\right)$$
$$E(X) = \left[M'(t)\right]_{t=0} = \frac{7}{2}.$$
$$E(X^2) = \left[M''(t)\right]_{t=0} = \frac{1}{6}\left[e^t + 4e^{2t} + 9e^{3t} + 16e^{4t} + 25e^{5t} + 36e^{6t}\right]_{t=0} = \frac{91}{6}$$
$$Var(X) = \frac{91}{6} - \frac{49}{4} = \frac{35}{12}.$$

→ Characteristic Function (CF) → CF of a RV X (discrete or continuous) is defined as $E(e^{i\omega x})$ & denoted as $\phi(\omega)$.

If X is DRV that can take values $x_1, x_2, \ldots$ & $P(X = x_r) = p_r$ then
$$\phi(\omega) = \sum_r e^{i\omega x_r} p_r$$
If X is CRV with density function $f(x)$, then
$$\phi(\omega) = \int_{-\infty}^{\infty} e^{i\omega x} f(x)\, dx.$$

Properties of CF → ① $\mu_n' =$ the coeff of $\frac{i^n \omega^n}{n!}$ in the expansion of $\phi(\omega)$ in series of ascending powers of $i\omega$.

② $$\mu_n' = \frac{1}{i^n}\left[\frac{d^n}{d\omega^n}\phi(\omega)\right]_{\omega=0}.$$

③ If CF of a RV X is $\phi_X(\omega)$ & if $Y = aX+b$ then
$$\phi_Y(\omega) = e^{ib\omega}\phi_X(a\omega).$$

④ If X & Y are independent RV then
$$\phi_{X+Y}(\omega) = \phi_X(\omega)\times\phi_Y(\omega).$$

Q. Find CF of the geometric distribution given by $P(X=r) = q^r p$, $r = 0,1,2,\dots$; $p+q=1$.
Find mean & variance.

Soln,
$$\phi(\omega) = \sum_{r=0}^{\infty} e^{i\omega r} p q^r = p\sum_{r=0}^{\infty}(q e^{i\omega})^r$$
$$= p(1-qe^{i\omega})^{-1}$$

$$\phi^{(1)}(\omega) = p(1-qe^{i\omega})^{-2}\, iq e^{i\omega}$$
$$\phi^{(2)}(\omega) = i^2 pq\left[2(1-qe^{i\omega})^{-3}qe^{i\omega} + (1-qe^{i\omega})^{-2}e^{i\omega}\right]$$
$$E(X) = \frac{1}{i}\phi^{(1)}(0) = \frac{q}{p} \quad \& \because p+q=1.$$
$$E(X^2) = \frac{1}{i^2}\phi^{(2)}(0) = \frac{q}{p^2}(1+q).$$
$$\mu_X = \frac{q}{p} \quad \& \quad \sigma_X^2 = \frac{q}{p^2}.$$
$\sigma_X^2 \to E(X^2) - [E(X)]^2$